

Question	Marking Guidance	Mark	Comments
09.1	1. Different parts/areas/amino acid sequences (of amyloid-precursor) protein; 2. Each enzyme is specific /fits/binds/ complementary to a different part of the APP;	2	1.Accept APP 2.Point 2 subsumes point 1 and is worth 2 marks total.
09.2	1. Peptide bond broken; 2. Using water;	2	Hydrolysis in stem
09.3	1. Mutations prevent production of enzyme(s)/functional enzyme; 2. (Increase in β -secretase) leads to faster/more β -amyloid production OR (Decrease in α -secretase) leads to more substrate for β -secretase; 3. (Leads to) more/greater plaque formation;	3	2.'This' must refer to α -secretase
09.4	1. (Inhibitor) binds to/blocks active site of β -secretase/enzyme; 2. Stops/reduces production of β -amyloid/plaque;	2	
09.5	1. Some β -amyloid required/needed (to prevent side effects) OR (Some) β -secretase needed; 2. Leads to build-up of amyloid-precursor protein (that causes harm) OR Too much product of α -secretase (causes harm);	1 max	1.Accept 'Both enzymes needed' 2. Accept build-up of substrate (leads to harm)